MSc Technologies and Analytics in Precision Medicine

Module: Advanced Data Analytics

Assignment 2

Name: Aminat Coster

Student Number: 25100668

March 2026

Contents

1 Introduction **2**

2 Methods **2**

2.1 Exploratory Data Analysis 2

2.2 Model Building 2

2.3 Classification Performance 3

2.4 ROC Curve and AUC Analysis 3

3 Results and Interpretation **3**

3.1 Descriptive Statistics 3

3.1.1 Interpretation 4

3.2 Model Building 4

3.2.1 Interpretation 5

3.3 Classification Performance 5

3.3.1 Interpretation 6

3.4 ROC Curve and AUC Analysis 7

3.4.1 Interpretation 8

4 Conclusion **8**

1 Introduction

This report aims to analyse the Pima Indians Diabetes dataset obtained from the *mlbench* package in R. The dataset, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases, contains diagnostic measurements from 768 Pima Indian women aged 21 years or older living near Phoenix, Arizona. This population has one of the highest reported rates of diabetes worldwide, making it a valuable cohort for investigating predictors of disease.

The dataset includes eight clinical and demographic predictors, such as plasma glucose concentration, blood pressure, body mass index (BMI), insulin levels, and age, alongside a binary outcome variable indicating diabetes status ('pos or 'neg'). Of the 768 individuals, 268 (35%) were diabetes-positive and 500 (65%) were diabetes-negative, reflecting a moderately imbalanced classification problem.

The aim of this analysis is to identify clinical and demographic factors associated with diabetes and to evaluate how accurately diabetes status can be predicted using logistic regression modelling. Odds ratios are estimated to provide clinically interpretable measures of association, and model performance is assessed using classification metrics and receiver operating characteristics (ROC) curve analysis. A logistic regression modelling approach was selected due to the binary nature of the outcome, allowing the probability of diabetes to be modelled appropriately.

2 Methods

2.1 Exploratory Data Analysis

Exploratory analysis summarised predictor distributions by diabetes status using means and standard deviations. Boxplots were generated to assess differences in distribution and potential separation between groups.

Problematic zero values in physiological variables (glucose, pressure, triceps, insulin and BMI) were converted to missing values (NA), as these measurements cannot be plausibly zero. Prior to model building, all observations containing missing values were removed resulting in a cleaned dataset comprising 392 individuals. This was done in order to ensure all models would be fitted to the same dataset, providing a consistent basis for comparison.

2.2 Model Building

Logistic regression was used to model diabetes status as a binary outcome. A full model including all predictors was initially fitted, followed by a reduced model including glucose, BMI (mass), and pedigree.

Models were compared using Akaike's Information Criterion (AIC) and a likelihood ratio test.

The model demonstrating better fit was retained as the final model. Odds ratios and 95% confidence intervals were obtained by exponentiating regression coefficients.

2.3 Classification Performance

Predicted probabilities from the final model were converted to binary classifications using a threshold of 0.5. A confusion matrix was constructed, and sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and accuracy were calculated.

2.4 ROC Curve and AUC Analysis

Receiver operating characteristic (ROC) curves were generated for both models using predicted probabilities. The area under the curve (AUC) and corresponding 95% confidence intervals were calculated to assess discrimination. AUCs were compared using DeLong's test. Youden's J statistic was used to identify an optimal classification threshold balancing sensitivity and specificity.

3 Results and Interpretation

3.1 Descriptive Statistics

Table 1 displays summary statistics stratified by diabetes status. The dataset comprises 500 individuals without diabetes and 268 individuals with diabetes. Mean glucose, BMI, age and pedigree values were higher in the diabetes-positive group. Differences in blood pressure and triceps thickness were minimal, with generally similar variability across groups.

Table 1: Summary Statistics (mean, SD) by Diabetes Status

Diabetes	Pregnant	Glucose	Pressure	Triceps	Insulin	Mass	Pedigree	Age	n
Neg	3.3	110.6	70.9	27.2	130.3	30.9	0.4	31.2	500
	(3.0)	(24.8)	(12.2)	(10.0)	(102.5)	(6.6)	(0.3)	(11.7)	
Pos	4.9	142.3	75.3	33.0	206.8	35.4	0.6	37.1	268
	(3.7)	(29.6)	(12.3)	(10.3)	(132.7)	(6.6)	(0.4)	(11.0)	

Figure 1 illustrates the distribution of the key predictor variables stratified by diabetes status. Glucose demonstrates the clearest separation between groups, with a substantial upward shift in median values and reduced overlap in the inter-quartile range among diabetes-positive individuals. BMI (mass) and age also show noticeable upward shifts in the positive group, although with greater overlap. In contrast, blood pressure and insulin display considerable overlap between groups, suggesting weaker unadjusted associations.

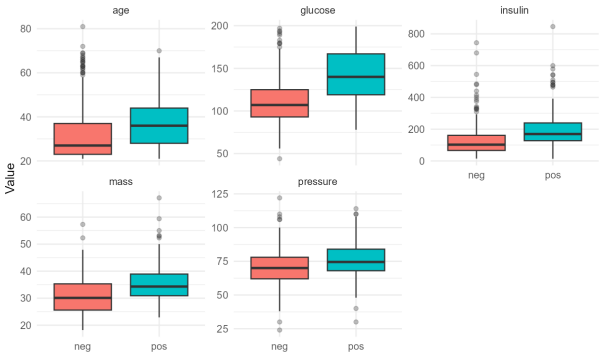

Figure 1: Distribution of Key Predictors between Groups

3.1.1 Interpretation

Overall, glucose appears most strongly associated with diabetes status, followed by BMI and age. The clearer separation observed for glucose suggests that it better distinguishes between individuals with and without diabetes, whereas the substantial overlap for pressure and insulin indicates weaker unadjusted associations. These exploratory findings provide an initial indication of which predictors may retain significance in subsequent modelling.

3.2 Model Building

Table 2 presents the odds ratios and 95% confidence intervals from the full logistic regression model including all predictors. Glucose, BMI (mass), and pedigree were positively associated with diabetes, with confidence intervals that do not include 1, indicating statistically significant effects. In contrast, pregnant, pressure, triceps, insulin, and age had confidence intervals that included 1, suggesting no statistically significant association with diabetes after adjustment for the other variables. Glucose demonstrated the most precise estimate, reflected by its relatively narrow confidence interval.

Table 2: Odds Ratios and 95% Confidence Intervals for the Full Model

Predictor	OR (95% CI)
Pregnant	1.09 (0.97–1.21)
Glucose	1.039 (1.028–1.051)
Pressure	0.999 (0.976–1.022)
Triceps	1.011 (0.978–1.046)
Insulin	0.999 (0.997–1.002)
Mass	1.073 (1.018–1.134)
Pedigree	3.130 (1.378–7.368)
Age	1.035 (0.999–1.074)

Table 3 displays the reduced model including only glucose, BMI, and pedigree. All retained predictors remained positively associated with diabetes and statistically significant, with confidence intervals excluding 1. The estimated effects are similar in magnitude to those observed in the full model, suggesting that removing the non-significant variables did not materially change the primary associations.

Table 3: Odds Ratios and 95% Confidence Intervals for the Reduced Model

Predictor	OR(95% CI)
Glucose	1.042 (1.032–1.052)
Mass	1.070 (1.031–1.114)
Pedigree	3.227 (1.454–7.383)

Table 4 compares the AIC values for the full and reduced models. The full model yielded a lower AIC (362.02) compared to the reduced model (371.70), suggesting that inclusion of additional predictors provides improved model fit despite increased complexity. This was supported by a likelihood ratio test comparing the two models (χ^2 = 19.68, p = 0.001), indicating that removal of the additional predictors significantly worsened model fit.

Table 4: AIC of Models

Model	AIC
Full Model	362.02
Reduced Model	371.70

3.2.1 Interpretation

Both the AIC comparison and likelihood ratio test indicated that the full model provided a better fit to the data, thus the full model was retained as the final model. A one-unit increase in plasma glucose concentration was associated with a 4% increase in the odds of diabetes, adjusting for other variables. Similarly, each one-unit increase in BMI corresponded to a 7% increase in the odds of diabetes. For each one-unit increase in the diabetes pedigree function, the odds of diabetes increase more than threefold. Given that pedigree values range approximately from 0 to 2.5, this reflects a substantial association.

3.3 Classification Performance

Table 5 presents the confusion matrix for the final model using a classification threshold of 0.5. Of the 262 individuals without diabetes, 233 were correctly classified as negative, while 29 were incorrectly classified as positive. Among the 130 individuals with diabetes, 74 were

correctly identified and 56 were misclassified as negative. Overall, the model demonstrates stronger performance in identifying non-diabetic individuals than in detecting diabetes cases at this threshold.

Table 5: Confusion Matrix (Threshold = 0.5)

	Actual	
Predicted	neg	pos
neg	233	56
pos	29	74

Table 6 summarises the classification performance of the final model using a threshold of 0.5. The model achieved a sensitivity of 56.9%, indicating that just over half of the individuals with diabetes were correctly identified. Specificity was substantially higher at 88.9%, demonstrating strong performance in correctly classifying non-diabetic individuals. The overall accuracy was 78.3%, although this may be influenced by the greater proportion of non-diabetic individuals in the dataset. The positive predictive value was 71.8%, while the negative predictive value was 80.6%, suggesting the model is more reliable at ruling out diabetes than detecting it. For screening purposes, the relatively low sensitivity is concerning, as a large proportion of diabetes cases would remain undetected at this threshold.

Table 6: Classification Performance Metrics (Threshold = 0.5)

Metric	Value
Sensitivity	56.9%
Specificity	88.9%
Positive Predictive Value (PPV)	71.8%
Negative Predictive Value (NPV)	80.6%
Accuracy	78.3%

3.3.1 Interpretation

The model demonstrates higher specificity than sensitivity, indicating stronger performance in identifying non-diabetic individuals than detecting diabetes cases. The relatively low sensitivity is clinically concerning, as false negatives represent missed diagnoses and delayed intervention, increasing the risk of long-term complications. In contrast, false positives may lead to temporary anxiety and additional confirmatory testing but are generally less harmful in a screening context. Given the importance of early detection, a lower classification threshold may be preferable to prioritise sensitivity, even at the expense of reduced specificity.

3.4 ROC Curve and AUC Analysis

Figure 2 illustrates the ROC curves for the full and reduced logistic regression models. Both curves lie well above the reference line representing random classification, indicating a good discriminatory performance. The curve for the full model is consistently closer to the upper-left corner, suggesting slightly improved discrimination compared to the reduced model.

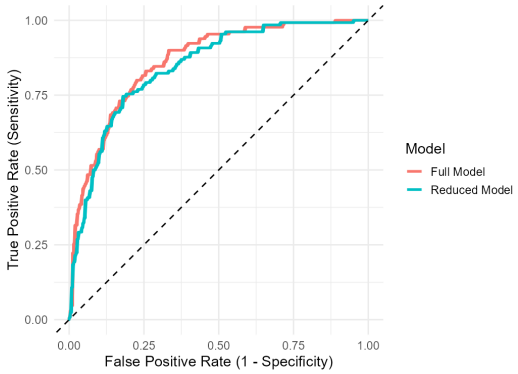

Figure 2: ROC Curves for Full and Reduced Models

Table 7 presents the AUC values and corresponding 95% confidence intervals for both models. The full model achieved an AUC of 0.8642, compared to 0.8448 for the reduced model, indicating slightly stronger overall discrimination. The confidence intervals for both models did not include 0.5, confirming performance was significantly better than chance. DeLong's test provided evidence of a slight difference between the AUCs ($p < 0.05$).

Table 7: AUC and 95% CIs for Full and Reduced Models

Model	AUC	95% CI
Full	0.8624	0.8251–0.8996
Reduced	0.8448	0.8050–0.8847

Table 8 summarises the optimal classification thresholds identifies using Youden's J statistic. For the full model, the optimal threshold was 0.29, yielding a sensitivity of 0.83 and a specificity of 0.74. The reduced model achieved an optimal threshold of 0.36, with sensitivity of 0.75 and specificity of 0.82. The full model demonstrates higher sensitivity at its optimal threshold, indicating improved detected of diabetes cases.

Table 8: Youden's J for Full and Reduced Models

Model	Threshold	Sensitivity	Specificity
Full	0.2853508	0.8307692	0.7442748
Reduced	0.3620955	0.7461538	0.8206107

3.4.1 Interpretation

An AUC of 0.86 indicates that the model has good ability to distinguish between individuals with and without diabetes. There is an 86% probability that the model will assign a higher predicted risk to a randomly selected individual with diabetes than to a randomly selected individual without diabetes. This suggests that the full model demonstrates strong ability to distinguish between diabetes-positive and diabetes-negative individuals.

4 Conclusion

This study examined which clinical demographic factors predict diabetes in Pima Indian women and evaluated how accurately diabetes status can be classified using these measurements. Logistic regression identified plasma glucose concentration, BMI, and diabetes pedigree function as the strongest predictors of diabetes, with pedigree demonstrating the largest estimated odds ratio and highlighting the importance of familial and genetic predisposition.

The strong effect observed for the diabetes pedigree function likely reflects inherited susceptibility to diabetes within the Pima Indian population, where familial clustering of type 2 diabetes has been widely documented ("(PDF) Diabetes Mellitus in the Pima Indians" 2026). Glucose and BMI were also positively associated with diabetes, reflecting key metabolic contributors to disease development. Elevated plasma glucose reflects impaired glucose regulation, while increased BMI is strongly associated with adiposity-driven insulin resistance, both of which are central mechanisms in the development of type 2 diabetes (Chandrasekaran and Weiskirchen 2024).

The full model demonstrated slightly better discrimination than the reduced model (AUC = 0.86), although the improvement was small. Overall discrimination was good; however, classification at the default threshold showed limited sensitivity, meaning a substantial proportion of diabetes cases would remain undetected. Lowering the threshold improved sensitivity and may be more appropriate in a screening context.

Several limitations should be acknowledged. Nearly half of the original dataset was excluded due to missing values, potentially introducing bias and reducing statistical power. Class imbalance and the relatively smaller number of diabetes-positive cases may have influenced model performance. Although the model shows good discriminatory ability, its classification performance and generalisability may be affected by data limitations.

In conclusion, glucose, BMI and familial risk are key predictors of diabetes in this population. While the model demonstrates good overall discrimination, sensitivity remains moderate, suggesting that threshold selection and data quality play important roles in clinical applicability.

References

Chandrasekaran, Preethi and Ralf Weiskirchen (Feb. 4, 2024). "The Role of Obesity in Type 2 Diabetes Mellitus—An Overview". In: *International Journal of Molecular Sciences* 25.3, p. 1882. ISSN: 1422-0067. DOI: 10.3390/ijms25031882. URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC10855901/ (visited on 03/07/2026).

"(PDF) Diabetes Mellitus in the Pima Indians" (2026). "Diabetes Mellitus in the Pima Indians: Genetic and Evolutionary Considerations". In: *ResearchGate* (). DOI: 10.1002/ajpa.1330620114. URL: https://www.researchgate.net/publication/16571215_Diabetes_Mellitus_in_the_Pima_Indians_Genetic_and_Evolutionary_Considerations (visited on 03/07/2026).